

Probability



Basic probability

- 1 Two fair six-sided dice are rolled. Find the probability their sum is 8.
-

Complement rule

- 2 If $P(E) = 0.35$, find $P(E')$.
-

Permutations and combinations

- 3 How many ways can a committee of 4 be chosen from 10 people?
- 4 How many ways can 2 officers (president, vice president — order matters) be chosen from 6 candidates?
- 5 Explain, using Theorem 6.1, how $\binom{6}{2}$ relates to $P(6, 2)$ from the previous question.
-

Independent events and the addition rule

- 6 Events A and B are independent, with $P(A) = 0.4$ and $P(B) = 0.5$. Find $P(A \text{ and } B)$.
- 7 A card is drawn from a standard 52-card deck. Find $P(\text{Queen or Spade})$.